

Day 1

ML application

- YT recommendation
- Spam detection in G-mail
- Voice Assistant
- Self Driving Car
- Face unlock in iPhones

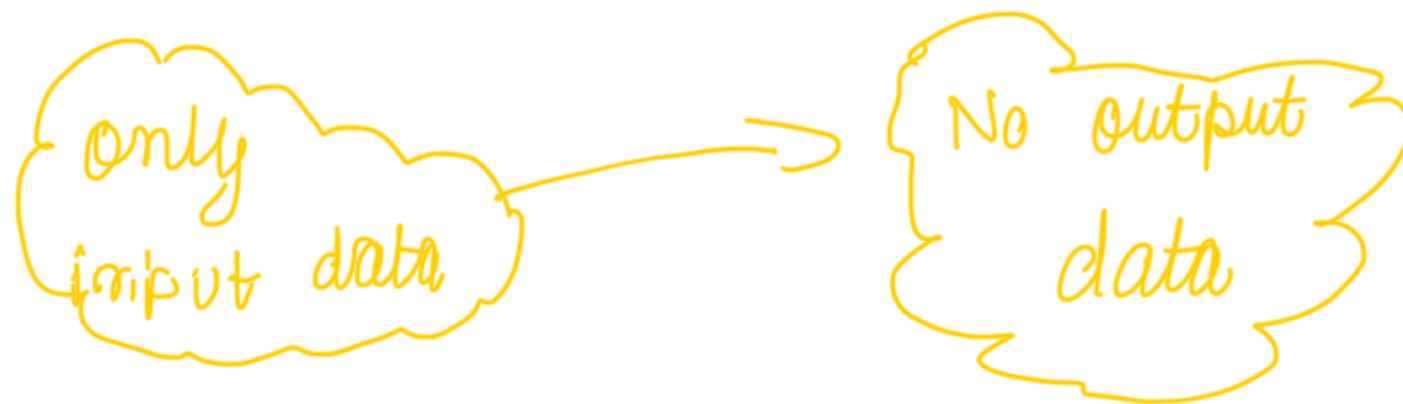
Types of ML

1) Supervised — we give inputs & answers & machine learns to give answers & predict it.

Algorithms

- Linear Regression
- Logistic Regression
- Decision Tree
- Naive Bayes

2) Unsupervised — we will just find patterns, group similarities, reduce complexity



3) Reinforcement Learning

It works on trial & error method.

We make our machine learn by trial & error

EDA

- 1) Problem definition
- 2) Data Collection
- 3) Exploratory Data Analysis
- 4) Data Preprocessing
- 5) Feature Selection
- 6) Split the dataset
- 7) Model Selection
- 8) Model Training

video 1

Model Tuning

a) Model evaluation

10) kly per parameter Tuning

11) Model Testion

EDA

{Exploratory Data Analysis}

It is used to

- understand it
- Discover pattern
- Spot anomalies
- generate insights
- and decide what to do next

Steps

- 1) View the data
- 2) Summary of stats
- 3) Value Counts → unique values, column repeat not there
- 4) Missing Value Analysis → gaps in Data
- 5) Visualization
- 6) Target Variable exploration

Importance

- can't clean data which you don't understand
- it helps you identify mistakes, biases or limitations
- It guides the dirⁿ of your data cleaning & feature engineering.
- it gives the audience & stakeholders a "story" or overview of what the data is telling you.